

Generative AI Fundamentals

1. What is Generative AI?

Generative AI refers to artificial intelligence systems that can generate new content based on patterns learned from training data. This includes text, images, code, audio, and more. Unlike discriminative models that classify data, generative models create new, original outputs.

2. Key Technologies

Transformers: Deep learning architecture using attention mechanisms for sequence-to-sequence tasks.

LLMs (Large Language Models): Neural networks trained on massive text data to understand and generate human language.

Diffusion Models: Generate images by iteratively denoising random noise conditioned on text prompts.

VAE (Variational Autoencoders): Learn compressed representations and generate new data similar to training data.

3. Training Approaches

Supervised Learning	Training on labeled data with input-output pairs
Unsupervised Learning	Learning patterns from unlabeled data
Reinforcement Learning	Learning through interaction and reward signals
Fine-tuning	Adapting pre-trained models to specific tasks

4. Popular GenAI Models

Model	Type	Use Case
GPT-4	LLM	Text generation, Q&A, code
Claude	LLM	Conversational AI, analysis
DALL-E	Diffusion	Image generation from text
Stable Diffusion	Diffusion	Image synthesis, editing
Whisper	Transformer	Speech-to-text recognition

5. Prompt Engineering

- Be specific and clear in your instructions
- Provide context and examples (few-shot learning)
- Use role-playing to guide model behavior
- Chain prompts for complex tasks
- Experiment with temperature and parameters

6. Ethical Considerations

- Bias in training data can propagate to outputs
- Hallucinations: Models may generate false information
- Copyright and intellectual property concerns
- Energy consumption and environmental impact
- Responsible deployment and monitoring required